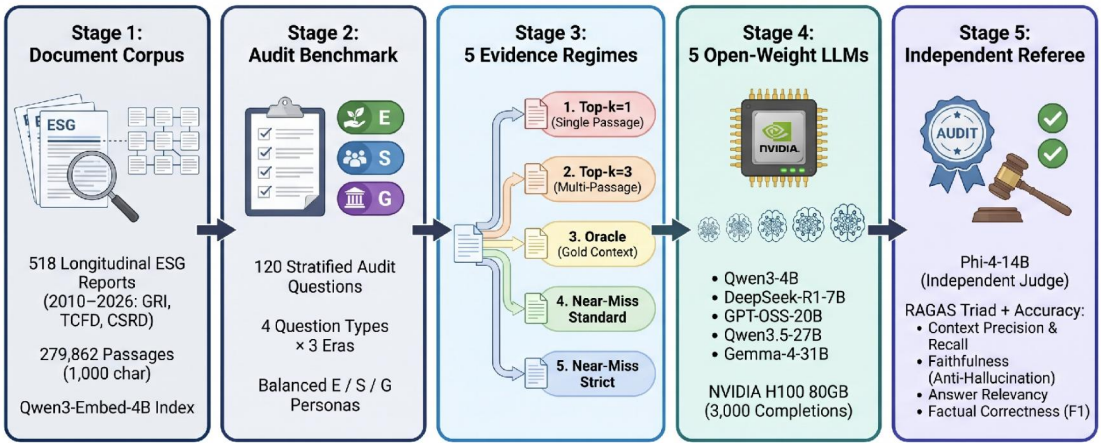

Stage 1:
Document Corpus
ESG
518 Longitudinal ESG Reports (2010–2026: GRI, TCFD, CSRD)
279,862 Passages (1,000 char)
Qwen3-Embed-4B Index
Stage 2:
Audit Benchmark
E
S
G
120 Stratified Audit Questions
4 Question Types × 3 Eras
Balanced E / S / G Personas
Stage 3:
5 Evidence Regimes
1. Top-k=1 (Single Passage)
2. Top-k=3 (Multi-Passage)
3. Oracle (Gold Context)
4. Near-Miss Standard
5. Near-Miss Strict
Stage 4:
5 Open-Weight LLMs
NVIDIA
• Qwen3-4B
• DeepSeek-R1-7B
• GPT-OSS-20B
• Qwen3.5-27B
• Gemma-4-31B
NVIDIA H100 80GB (3,000 Completions)
Stage 5:
Independent Referee
AUDIT
Phi-4-14B (Independent Judge)
RAGAS Triad + Accuracy:
• Context Precision & Recall
• Faithfulness (Anti-Hallucination)
• Answer Relevancy
• Factual Correctness (F1)